ENVIRONMENTAL DATA SCIENCE

APPLICATION PAPER

Convolutional neural network for the prediction of Sargassum seaweed beachings in Guadeloupe

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Abstract

Impact Statement

For several years, the archipelago of Guadeloupe has faced massive arrivals of sargassum seaweed, which have a lasting impact on its coastline, economy, and biodiversity. While deep learning methods are now delivering significant results across a wide range of complex fields, their application to this local environmental challenge represents a major technological opportunity. We have therefore undertaken the development of a sargassum arrival prediction tool specifically tailored to the Guadeloupean coastline. This study is based on the exploitation of satellite data coupled with the analysis of key climatic variables, such as ocean currents and winds. By using deep neural networks capable of processing this multitude of factors, our approach identifies the dynamics of seaweed movement with increased precision. Through this strategic steering instrument, the project aims to consolidate territorial organization for more effective sargassum management. It thus offers a concrete solution to sustainably protect the coastline and local populations in the face of this significant environmental challenge.

In recent years, deep learning has transformed data processing, justifying its use in the current study. These models consist of multiple layers that calibrate weights to optimise accuracy. By mimicking neural processes, these networks facilitate high-performance predictive modelling. This study focused on the application of convolutional neural networks for forecasting Sargassum seaweed arrivals in Guadeloupe, using varied climatic data such as winds, surface currents, satellite images and observations of seaweed arrivals on the Guadeloupe coast. We tested two different approaches : classification models for forecasting up to 14 days and zone regression models for forecasting up to 14 days, also indicating a probability of algae arrival. The results showed that the classification models achieved a mean accuracy of 94.75%, a mean specificity of 97.42%, and a mean sensitivity of 98.78%. Zone regression models achieved a mean accuracy of around 90%, a mean specificity of

around 97%, and a mean sensitivity of around 85%, with a mean MAE of 0.13 and a mean RMSE of 0.18. These models perform better than the decision tree proposed in the state of the art for the same problem.

1. Introduction

Since 2011, massive Sargassum algae beaching have been observed on the coasts of the Lesser Antilles (LA), mainly those facing east and southeast (Wang and Hu, 2016; Bernard et al., 2022). The LA received large quantities of algae on the windward Atlantic coast, while the leeward Caribbean coastal areas remained little affected.

In terms of frequency and intensity, these beachings can now be considered a new natural hazard for the Caribbean islands and the American coastline. While it has been demonstrated that Sargassum algae provide habitats and shelter for various organisms in a structurally sterile ocean ecosystem, the beachings of the last decade have induced health risks for the population and had considerable socioeconomic impacts. For example, if we consider the French West Indies, the Guadeloupe archipelago and Martinique, the results are as follows.

- Apart from 2013, the recent inflow of Sargassum rafts onto the coasts of Guadeloupe and Martinique, although irregular, has not ceased since 2011, reaching a paroxysm in 2015 (Florenne et al., 2016; Berline et al., 2020). State services have estimated that the volumes beached on the coasts were on the order of 1.5 million m^3 from October 2014 to October 2015 in Guadeloupe (Florenne et al., 2016). Only a third of this volume could be collected by the authorities. The particularity and difficulty lay in the fact that a large part of the coastline and beaching volume remained inaccessible to currently proven techniques and at costs that are now bearable.
- This has had an impact on human health and ecosystems, because in shallow, small bays, accumulated algae degrade by fermentation and emit chemical compounds such as hydrogen sulphide (H_2S) and ammonia (NH_3) (Resiere et al., 2018).
- A survey conducted by the organizations responsible for socioeconomic development estimated that the decline in tourism resulted in an economic loss of USD 5.5 million for the first half of 2015 (https://eos.org/features/Sargassum-watch-warns-of-incoming-seaweed)

The volumes to be collected were considerable compared to the size of these islands (<1200 km^2 each) and the vulnerability of these territories. This new phenomenon has raised several scientific questions relating to Sargassum rafts, such as their transports, origins, and the sources of nutrients promoting their growth, especially the physical factors that led to their occurrence and their development in the tropical and equatorial Atlantic.

The roles of nutrient input and surface current transport were estimated using satellite images of the ocean, reanalysed wind and surface current data, Sargassum volume records collected on ships and numerical models. Several authors have contributed to the understanding of the mechanisms and physicochemical processes governing the phenomenon (Johns et al., 2020). Operational systems have

been developed, such as the satellite-based Sargassum Watch System SaWS (https://optics.marine.usf.edu/projects/SaWS.html). They provide a temporal and spatial assessment of annual seasonal increases and decreases in the amount of Sargassum algae over large areas of the tropical Atlantic and Caribbean (Wang et al., 2019).

Tropical Atlantic currents and winds gather and transport these algae seasonally to the Caribbean (Brooks et al., 2018; Cuevas et al., 2018; Johns et al., 2020; Jouanno et al., 2021). Modelling studies have mainly focused on the transport properties of Sargassum rafts by ocean currents (Wang et al., 2019; Berline et al., 2020). These analyses have been extended to highlight the abnormal transport of Sargassum rafts related to certain atmospheric phenomena (Johns et al., 2020). These various studies encourage the use of surface current and wind data to build a model for predicting Sargassum algae beaching.

In their previous work, Bernard et al., (2022) used a combination of satellite-based floating algae index (AFAI) (Wang and Hu, 2016) with the Hybrid Coordinate Ocean Model (HYCOM) surface current forecast data to predict short-term Sargassum beaching for Guadeloupe. (Trinanes et al., 2021) presented the Sargassum inundation reports (SIRs), a product based on satellite observations to predict Sargassum coastal inundation potential weekly throughout the Caribbean Sea region, the Gulf of Mexico, and extending to the east coast of Florida and the Bahamas. As described by Trinanes et al., (2021), the SIR algorithm uses the floating algae density values within 50 km of each coastal pixel to predict three inundation potential levels (low, medium, and high). This algorithm does not include ocean currents, winds, and waves, which may modify the movement of Sargassum.

The implementation of methods based on several independent data sets has led to the production of scientific knowledge and even to the development of largescale forecasting systems. Bernard et al. (2022) used a clustering method combined with an expert metric (Biabiany et al., 2020; 2026) to identify current patterns. These patterns were coupled with wind reanalysis, satellite-based Sargassum offshore abundance and past observed beachings, all of which were integrated into a decision tree model. This model produces a probability of algae beaching within 7 days, with an accuracy of 73%. Decision trees are not very recent and could be replaced by more powerful artificial intelligence methods. Indeed, various studies have used neural network methods on climate data, such as multi-layer perceptron (Hunasigi et al., 2023), self-organizing map (Dey et al., 2023) and convolutional neural network (Weyn et al., 2019; Weyn et al., 2020; Chattopadhyay et al., 2020 and Rasp and Thuerey 2021).

In the following, methods using neural networks and the data used by Bernard et al., (2022), will be presented. The aim is to build a model for predicting Sargassum beaching on the Guadeloupe coast. All the methods and data will be presented in detail in section 2 and the results of the models built will be presented in section 3.

2. Materials and methods

This study focuses on the Lesser Antilles arc (*cf.* Fig. 1A), and the data used covers the period from January 1, 2019, to December 31, 2023. However, some daily data from 2018 have been added as input to our methods to improve automatic analysis. The datasets used take the form of matrices of numerical values, providing geolocalized information on physical and environmental parameters. The data sources used are presented in the following subsection.

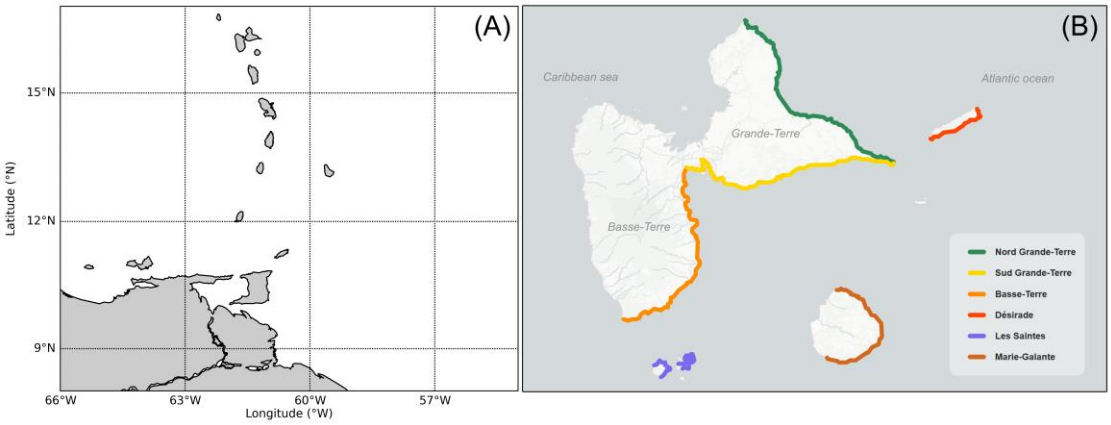

Fig. 1. (A) Study area: Lesser Antilles arc from 55- 66°W and 8-17°N; (B) Guadeloupe coastlines: Nord Grande-Terre (NGT) in green line, Sud Grande-Terre (SGT) in yellow line, Basse-Terre (BT) in orange line, Désirade in red line, Les Saintes (LS) in purple line and Marie-Galante (MG) in brown line.

Before being used in automatic analysis methods, each dataset is subjected to internal normalization (*cf.* Eq. 1) to optimize the operation of the algorithms, as follows:

$$d' = \frac{(d - \min(D))}{\max(D) - \min(D)} \; where \; d \in D \quad \textbf{(1)}$$

where D represents a dataset (i.e., all days within the study period), d represents a data matrix for one day and d′ the latter after normalization. This operation is performed for each day d ∈ D and for each dataset. For processing purposes, the different datasets were interpolated to a common resolution of 0.08° in longitude and 0.04° in latitude using a bilinear method. The 12:00 Coordinated Universal Time (UTC) forecast data (corresponding to 08:00 AM local time in Guadeloupe) was selected to ensure data availability in the context of daily operational forecasting.

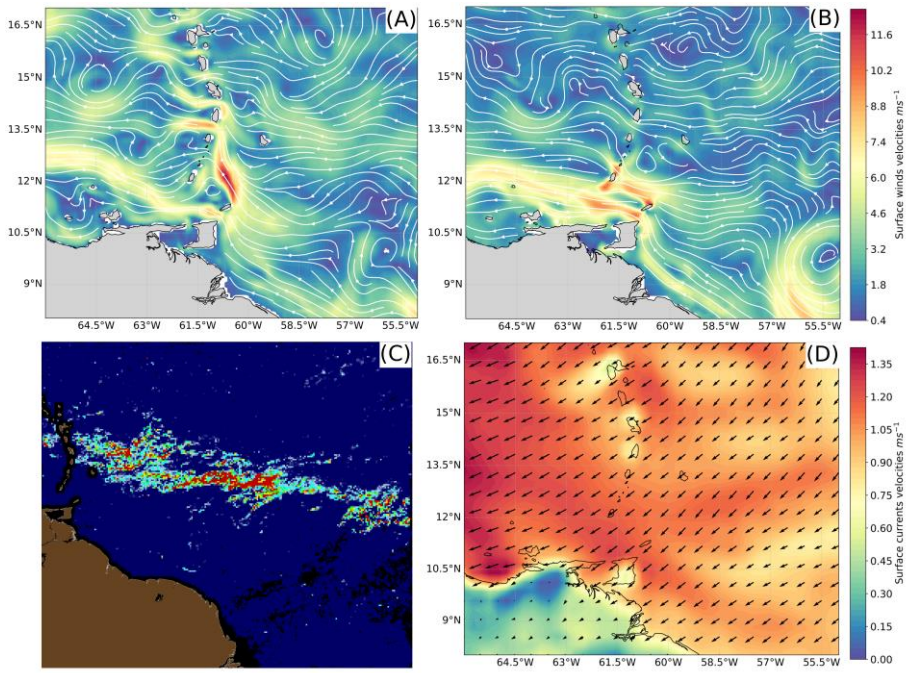

Fig. 2. Spatial maps of the different predictors for January 1st, 2019: (A) HYCOM surface current; (B) Mercator surface current; (C) FA-Density satellite image; (D) ERA-5 surface winds.

2.1. HYCOM surface current dataset

Daily surface current components from the 41-layer Hybrid Coordinate Ocean Model (HYCOM), at 1/12° resolution, global analysis (HYCOM GLBy0.08 version, available at: https://www.hycom.org/data/glby0pt08/expt-93pt0), were examined. The HYCOM surface forcing, including 10m wind velocities, is extracted from Climate Forecast System Version 2 (CFSv2) (*cf.* Fig. 2A). The Navy Coupled Ocean Data Assimilation (NCODA) system is used to assimilate available observational data: satellite altimeter sea surface height, satellite and in situ sea surface temperature, temperature vertical profiles, and salinity vertical profiles. The HYCOM GLBy0.08 grid resolution is 0.08° in longitude and 0.04° in latitude. To perform the present study, the native HYCOM fields were normalized.

2.2. Mercator surface current dataset

Daily surface current component data sourced from the Mercator PSY4V3R1 analysis system were utilized. This dataset features a 1/12° resolution with 50-layer 3D analysis and a uniform 0.08° resolution in latitude and longitude. For this study, the data were also normalized and interpolated (*cf.* Fig. 2B).

2.3. ERA-5 dataset: surface winds

Daily surface wind data were also considered, as they have an impact on the transport of Sargassum in the water. Surface wind components with a horizontal resolution of 31km from the ERA5 reanalysis model (available here: https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-pressure-

levels?tab=overview), were used (*cf.* Fig. 2D). The data were normalized (*cf.* Equation 1) and interpolated using a bilinear method, so that the data had the same dimensions as the HYCOM surface current data.

2.4. Satellite-based offshore abundance of Sargassum

We also consider images from several satellites. Floating algae has a particular wavelength, and it is this wavelength that is used to obtain the algae flotation index anomaly (AFAI). However, the heavy cloud cover in the Lesser Antilles leads to temporary unavailable data on certain days. For this reason, we use the floating algae density index (FA-Density) (*cf.* Fig. 2C), which is obtained from 7 consecutive days of AFAI data. The more algae pass through a zone, the higher the FA-Density. This process gives us images that provide information on the position and density of algae with a resolution of 1°. These data have been normalized and interpolated, to have the same dimensions as the HYCOM data.

2.5. Beaching observational data

Observations of Sargassum beachings on the Guadeloupe coast, provided by the Direction de l'Environnement, de l'Aménagement et du Logement (DEAL) de Guadeloupe from 2017 to 2019 and by Météo France from 2020 onwards, have been used to create the labels. These labels are the responses that will be given to the neural networks so that they can train themselves.

Two labeling methods were used: for the first method, beaching was considered to have occurred as soon as the probability of beachings in the observation data was high. The result is a matrix with a size corresponding to the number of days between January 1, 2019, and December 31, 2023 (*i.e.,* 1826 days). The label will be 0 if there are no beachings and 1 if there are. Of the 1826 days analyzed, 277 days of beachings are observed.

For the second method, only Météo France observations were considered. The bulletins published by Météo France contain the probability of Sargassum algae beachings for Nord Grande-Terre (NG), Sud Grande-Terre (SGT), Basse-Terre (BT), Désirade, Les Saintes (LS) and Marie-Galante (MG) (*cf.* Fig. 1B). Forecasts are valid from the day of issue of the bulletin and for the following 3 days. For each bulletin we had available, the probability of beachings for each zone was assigned to the date on which the bulletin was published and to the three following days. The other dates were initialized to 0 for all zones. The result is a matrix with the following dimensions (1461, 6), where 1461 is the number of days in the period from January 1, 2020, to December 31, 2023, and 6 is the number of zones. Over this period, there are 1027 days where there is a risk of beachings and 434 where there is no risk.

2.6. Dataset creation

Convolutional neural networks (CNN) can process input data tensors or handling input data with variable depth. In the case of images, a CNN processes an input with a depth comprising three colors of channels (Red, Green, Blue) (Taye., 2023). So, we're going to build our own tensors from the data presented in the previous section. In this case, the input data will have a seven-level depth. These are

seven separate parameters: the U_0 and V_0 components of the surface currents from the HYCOM and Mercator models, the U and V components of the wind, and finally, satellite measurements of Sargassum density. To build the data tensors, we first resized each dataset so that they all had the same dimensions: those of the HYCOM model. We then normalized each dataset and stacked the parameters for each day one after the other (cf. Fig. 3). We thus end up with a tensor of dimensions (1826, 7×n, 226, 138) Here, each element of the tensor represents one day of the study period, n represents the number of input days supply to the neural network, and each day consists of 7 variables. This tensor represents the dataset that will be used in the two approaches presented in the following sections.

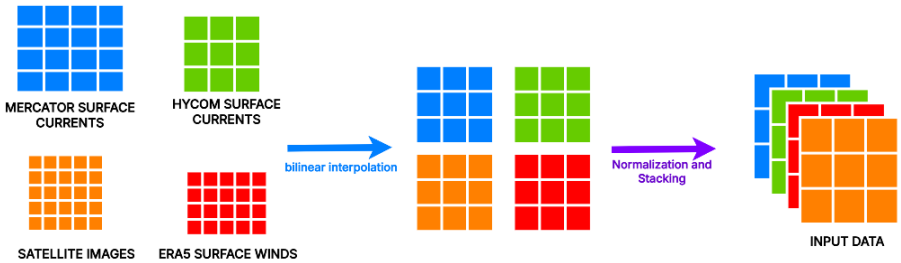

Fig. 3. Data preprocessing: The satellite images and the components U, V, U_0, V_0 are resized, normalized, and the stacked to a tensor (1826, 7×*n*, 226,138).

2.7. Classification models

Before building a new network, the possibility of transfer learning was mentioned. This involves using a model that has already been trained for a task similar to the one we want to perform and adapting it to our problem. This method reduces the immense time and resource demands of training most neural network models. Pre-trained models exist for solving problems related to Sargassum algae (Laval et al., 2023; Wang and Hu, 2021), however, none of these models use the data presented in this study and none are focused on the area presented. For this reason, the models presented below are built from scratch. To build our model for forecasting Sargassum algae arrivals, we began by designing a classification model, using convolutional neural networks. Our objective is to assign a class to every entry in the dataset.

In this study, two classes are considered: days with beachings and days without beachings. As mentioned in section 2.5, out of 1826 days there are only 277 days with beachings, and therefore 1549 days with no beachings. We note a class imbalance, which could affect the ability of our models to assign the right class to each day, due to the low number of examples of days with beachings. To address this issue, the class representing beaching days was oversampled. This was achieved by duplicating samples of beaching events until class parity was reached, resulting in a newly balanced dataset. While other techniques such as SMOTE or data augmentation were considered, they were dismissed as they could alter the physical properties and spatial coherence of the data, elements we specifically aimed to preserve. This dataset was then split into a training dataset and a test dataset using a 70:30 ratio, resulting in 2,168 training samples and 930 testing samples. Both the training and testing sets maintain a perfect parity between beaching and non-beaching events.

Each CNN consists of five feature extraction blocks followed by a fully connected network, which is in fact a multi-layer perceptron. Each feature extraction block consists of a convolutional layer using kernels of size 3x3, followed by a batch normalization (BN) and a max pooling layer of size 2x2. Thus, at the output of the pooling layer, several feature maps (FMs) are extracted. The first two blocks generate 64FM, 128FM are produced at the next two layers, and finally the last block extracts 256FM. The feature maps produced in the last block are then vectorized, to be passed as input to the fully connected part, which handles classification (*cf.* Fig. 4).

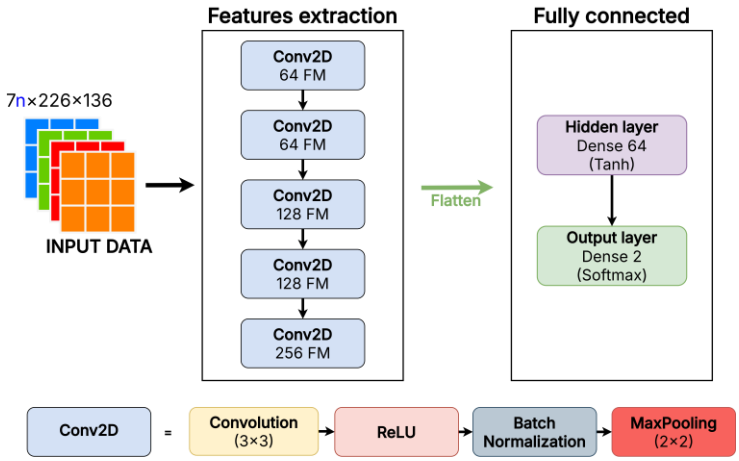

Fig. 4. CNN based architecture for classification; *n* corresponds to the number of days provided as input.

The class assignment part has a single hidden layer, containing 64 neurons. The output layer is made up of 2 neurons, corresponding respectively to the class of days without beachings and that of days with beachings. Finally, the ReLU activation function is used for the convolutional layers. For the classification part, the hyperbolic tangent function is applied to the hidden layer, while the softmax function is used for the output layer. The results obtained are detailed in Section 3.1.

2.8. Regression models

Still with the aim of predicting the arrival of sargassum seaweed, a second approach was tested. This approach consists of a model with 6 neurons on its output layer, each of which corresponds to a Guadeloupe coastline (*cf.* Fig. 1B). The output value indicates the probability of Sargassum arriving on the coast concerned. These models will be referred to as "*ZonalReg*" models. For these models, oversampling was not used because the task shifted from a classification problem to a regression problem, where the class imbalance issue is more negligible. Furthermore, as previously mentioned, the Météo France bulletins provided beaching probabilities for each zone over a four-day period. The dataset for these models was also split using a 70:30 ratio, resulting in a training set of 1,022 examples and a test set of 439 examples.

The architecture of the feature extraction part is almost identical to that presented for the classification model. The only difference is that the Batch Normalization and Max Pooling layers in the Conv2D block have been inverted. The number of neurons on the output layer of these models changes since the

aim is not to assign a class to each entry, but to provide a probability of beachings per zone. For the same reason, the activation function used on the output layer is no longer the softmax function but the sigmoid function. Finally, the activation function used on the hidden layer of the fully connected part is the ReLU function (*cf.* Fig. 5). The results of experiments with these two models will be detailed in section 3.2.

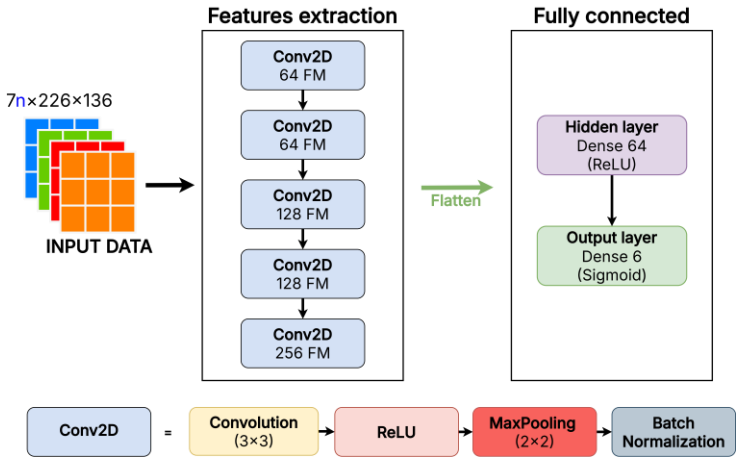

Fig. 5. *ZonalReg* model architecture where n corresponds to the number of days provided as input.

3. Discussions

To achieve optimal performance, extensive experiments were conducted by varying key hyperparameters, including the learning rate, batch size, the number of days provided as input to the network, and the forecast lead time. The models were implemented in Python using the TensorFlow library and the Keras API. Furthermore, the training loops were executed on the C3I high-performance computing cluster of the University of the Antilles, enabling the development of models capable of predicting beaching events up to 14 days in advance. Among the configurations tested, the most effective models were those utilizing a single day of input data single day of input data, represented as tensors of 7 parameters.

To evaluate model performance, distinct metrics were employed for both classification and regression tasks. Classification results were assessed based on accuracy, sensitivity (recall), and specificity (*cf.* Appendix A). For the regression analysis, the Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE) (*cf.* Appendix A) were utilized to quantify predictive precision. Furthermore, to enable a direct comparison between the two approaches, classification metrics were also calculated for the regression models. This was achieved by applying a threshold of 0.5 to the continuous outputs, thereby transforming the regression results into binary classifications. Beyond these performance metrics, identifying the specific contribution of each input variable to the model's predictions remains a key objective for future analysis. To determine which predictor has the greatest impact on the forecasts, an interpretability study using Integrated Gradients (Sundararajan et al., 2017; Singh et al., 2023) could be

conducted. This method would provide a more transparent understanding of the underlying physical mechanisms captured by the network, moving beyond simple scores to explain how specific parameters influence beaching events.

Due to the method used for class rebalancing, which involves duplicating days with beachings, it is possible that model performance may be overestimated. For the sake of clarity, only a selection of the most pertinent results is presented in this report.

3.1 Classification models

Table 1 (*cf.* Appendix B) details the performance of the 15 classification models from D+0 to D+14, with best and worst scores highlighted in green and red, respectively. Performance is consistently high, with average accuracy, specificity, and sensitivity reaching 94.75%, 97.42%, and 98.79%. The D+14 model achieves peak accuracy and specificity, while the highest sensitivity is shared by the D+0 and D+2 models. Even the minimum scores remain robust, exceeding 93% across all metrics. The high specificity reflects the model's effective handling of class imbalance, accurately identifying the predominant "no-beaching" days. Simultaneously, the strong sensitivity scores demonstrate the model's capability to correctly detect actual beaching events. Overall, these results indicate that the networks successfully capture the underlying patterns for both classes across the entire 14-day forecast horizon.

3.2 *ZonalReg* models

Tables 2 and 3 (*cf.* Appendix B) show the evaluation of metrics for the 15 zonal regression models. The regression analysis reveals that both MAE and RMSE values exhibit minimal variation across the different models, maintaining consistently low levels with an average error of 0.13 and 0.18, respectively. This remarkable stability in error margins indicates that the network's predictive precision remains robust throughout the entire forecasting range. Consequently, the models can deliver stable and accurate predictions up to a 14-day lead time without significant degradation in performance.

In Table 3, the optimal scores for each metric are highlighted in green, while the lowest performances are indicated in red. The scores for accuracy, sensitivity, and specificity derived from these regression models demonstrate strong results. Specifically, we observed a maximum (minimum) accuracy of 92.74% (88.16%), sensitivity of 89.89% (76.97%), and specificity of 98.96% (95.80%). Although these results are high, they remain slightly below those of the dedicated classification models. This difference can be explained by the application of a fixed threshold to the continuous outputs; unlike classification models, this method leaves less margin for error.

Despite slightly lower classification scores, the regression approach provides superior geographical precision. While we initially considered a multi-model classification approach consisting of six classifiers, one for each zone, to determine whether or not a zone would be affected, we chose to develop these regression models to provide the beaching probability for each zone. This allowed us to quantify the level to which each area is affected rather than simply stating if an impact occurs. Consequently,

the regression models deliver specific predictions for each coastal zone, whereas classification models only offer a single global output for the entire coastline.

4. Conclusion

The objective of this study was to build a convolutional neural network model capable of predicting the arrival of Sargassum algae on the coasts of Guadeloupe. This model was intended to use data on winds, surface currents, satellite images, as well as observations of beachings and arrivals of these algae on the shores. To achieve this goal, several classification and regression models were proposed. Both approaches demonstrated superior performance compared to traditional decision tree proposed in a previous study, which achieved scores of 70.10% for accuracy, 73.10% for sensitivity, and 69.20% for specificity. In comparison, the proposed CNN models reached significantly higher results, with accuracy up to 95.70%, sensitivity of 99.64% and specificity of 98.96%, the regression models also achieve an average MAE of 0.13 and an average RMSE of 0.18 confirming that deep learning architectures are better suited for capturing these complex dynamics over a 14-day lead time.

A significant advantage of this study lies in the geographical precision of regression models. Although classification models achieved higher metrics, the regression approach is more relevant for precise monitoring. Despite a slight performance gap which can be explained using a fixed threshold, these models deliver specific predictions for each coastal zone rather than a single global output.

This study currently demonstrates that there is a significant predictability within the data that can be effectively learned by the models. However, as these models were trained on reanalysis data, it would be valuable to investigate whether similar performance levels are maintained when using real-time operational data. Future work could focus on improving regression scores to bridge the performance gap, notably by identifying an optimal threshold through ROC curve analysis for a more balanced comparison. Furthermore, conducting an analysis with Integrated Gradients could help determine which predictors have the greatest impact on the forecasts. Furthermore, the robust performance observed even at a 14-day lead time encourages us to extend the forecasting horizon in subsequent studies to provide earlier warnings for coastal management.

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Appendix A: Evaluation metrics

$$Accuracy = \frac{1}{n}\sum_{i=1}^{n}(\widehat{y_i} = y_i) \quad (2)$$

where n is the size of the dataset, $\hat{y}_i$ is the prediction made by the model and y_i is the expected response

$$Sensitivity = \frac{True\ positive}{(True\ positive + False\ negative)} \quad (3)$$

$$Specificity = \frac{True\ negative}{(True\ Negative + False\ positive)} \quad (4)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \quad (6)$$

Appendix B: Score tables for the different models.

Table 1. Evaluation of classification models: Accuracy (%), Specificity (%) and Sensitivity (%), D to D+14.

Day	Accuracy (%)	Specificity (%)	Sensitivity (%)
+ 0	94.30	96.71	99.64
+ 1	94.41	97.42	98.56
+ 2	94.30	96.64	99.64
+ 3	95.05	97.55	98.19
+ 4	94.95	97.61	98.19
+ 5	94.73	97.61	98.55
+ 6	93.98	97.48	98.55
+ 7	94.52	97.61	98.55
+ 8	94.73	97.10	99.28
+ 9	95.38	97.48	99.28
+ 10	95.48	97.61	99.28
+ 11	95.48	98.00	98.55
+ 12	93.44	97.03	98.19
+ 13	94.84	97.35	98.55
+ 14	95.70	98.06	98.91

Table 2. Evaluation of zonal regression models: MAE and RMSE, D to D+14.

Day	MAE	RMSE
+ 0	0.13	0.18
+1	0.13	0.17
+ 2	0.13	0.17
+ 3	0.14	0.18
+ 4	0.13	0.18
+ 5	0.14	0.18
+ 6	0.14	0.19
+ 7	0.14	0.18
+ 8	0.13	0.18
+ 9	0.13	0.18
+ 10	0.14	0.19
+ 11	0.14	0.19
+ 12	0.13	0.18
+ 13	0.14	0.19
+14	0.13	0.18

Table 3. Evaluation of zonal regression models: Precision (%), Specificity (%) and Sensitivity (%), from D to D+14, for North Grande-Terre (NGT), South Grande-Terre (SGT), la Désirade, la Basse-Terre (BT), Les Saintes (LS) and Marie-Galante (MG), green highlighted values are the optimal scores for each metric and the worst are in red.

Day	Metrics	NGT	SGT	Désirade	BT	LS	MG	Average
+0	**Accuracy (%)**	90.28	90.42	90.97	90.01	89.46	90.21	90.22
	Specificity (%)	97.25	98.80	98.33	97.13	96.62	97.55	97.61
	Sensitivity (%)	84.21	82.31	87.76	87.73	85.32	86.28	85.60
+ 1	**Accuracy (%)**	90.21	91.79	92.27	90.28	90.49	90.90	90.99
	Specificity (%)	97.63	98.50	98.71	98.96	97.95	98.48	98.37
	Sensitivity (%)	87.50	86.15	87.76	89.17	88.74	87.00	87.72
+ 2	**Accuracy (%)**	90.83	91.38	91.10	90.42	91.10	90.08	90.82
	Specificity (%)	98.09	98.35	97.72	97.38	97.60	97.55	97.78
	Sensitivity (%)	80.92	89.23	88.44	89.53	87.37	89.89	87.56
+ 3	**Accuracy (%)**	90.14	91.03	91.03	90.14	88.84	88.91	90.02
	Specificity (%)	98.09	98.42	97.72	97.89	97.17	96.71	97.67
	Sensitivity (%)	78.95	84.62	85.71	84.84	83.62	88.09	84.30
+ 4	**Accuracy (%)**	91.51	91.1	90.35	90.69	89.53	88.57	90.29
	Specificity (%)	98.47	98.05	97.95	97.30	97.69	97.38	97.81
	Sensitivity (%)	80.26	83.85	83.67	86.64	87.03	85.56	84.5
+ 5	**Accuracy (%)**	91.72	90.69	89.39	90.08	89.32	89.39	90.10
	Specificity (%)	97.86	97.15	97.79	97.47	95.80	97.30	97.23
	Sensitivity (%)	81.58	86.92	85.03	85.92	88.05	88.45	85.99
+ 6	**Accuracy (%)**	90.97	91.58	90.9	89.53	88.77	88.43	90.03
	Specificity (%)	98.17	98.42	97.79	98.14	97.26	96.96	97.79
	Sensitivity (%)	85.53	87.69	80.95	83.39	83.28	84.48	84.22
+ 7	**Accuracy (%)**	90.42	91.58	91.85	90.42	89.32	89.66	90.54
	Specificity (%)	96.33	97.97	97.49	97.21	96.58	97.13	97.12
	Sensitivity (%)	82.89	87.69	84.35	89.89	84.98	88.45	86.38
+ 8	**Accuracy (%)**	92.74	91.58	90.49	90.01	89.53	88.71	90.48
	Specificity (%)	98.17	97.97	97.34	96.28	97.35	96.88	97.33
	Sensitivity (%)	88.16	87.69	88.44	88.81	86.69	86.28	86.65
+ 9	**Accuracy (%)**	89.66	89.94	90.9	91.38	89.6	89.25	90.12
	Specificity (%)	96.87	98.05	97.26	97.38	97.26	97.30	97.33
	Sensitivity (%)	84.21	83.85	87.76	88.45	85.67	87.73	86.28
+ 10	**Accuracy (%)**	90.14	89.87	89.73	89.46	90.01	88.16	89.56
	Specificity (%)	98.09	98.12	97.87	96.96	97.17	97.64	97.64
	Sensitivity (%)	85.53	81.54	82.31	86.64	82.59	80.87	83.25
+ 11	**Accuracy (%)**	91.79	91.65	90.90	89.46	89.46	89.6	90.48
	Specificity (%)	98.70	98.50	97.34	97.21	97.52	97.55	97.80
	Sensitivity (%)	82.24	80.00	83.67	84.48	83.96	82.67	82.84
+ 12	**Accuracy (%)**	90.76	90.69	90.76	90.69	90.08	88.64	90.27
	Specificity (%)	98.17	98.20	98.02	97.64	97.17	96.88	97.68
	Sensitivity (%)	82.24	81.54	80.95	85.56	86.01	85.56	83.64
+ 13	**Accuracy (%)**	89.80	91.65	91.72	90.35	89.12	89.05	90.28
	Specificity (%)	98.17	98.20	98.02	97.47	96.58	96.54	97.50
	Sensitivity (%)	76.97	86.92	85.03	87.73	84.98	88.81	85.07
+ 14	**Accuracy (%)**	91.72	91.72	91.10	89.94	89.73	89.80	90.67
	Specificity (%)	98.01	98.27	97.87	96.71	96.66	96.62	97.36
	Sensitivity (%)	78.95	87.69	84.35	89.17	86.35	87.00	85.58

Average	**Accuracy (%)**	90.85	91.10	90.90	90.19	89.62	89.29	-
	Specificity (%)	97.87	98.20	97.81	97.41	97.09	97.23	-
	Sensitivity (%)	82.68	84.77	85.08	86.89	85.64	86.47	-